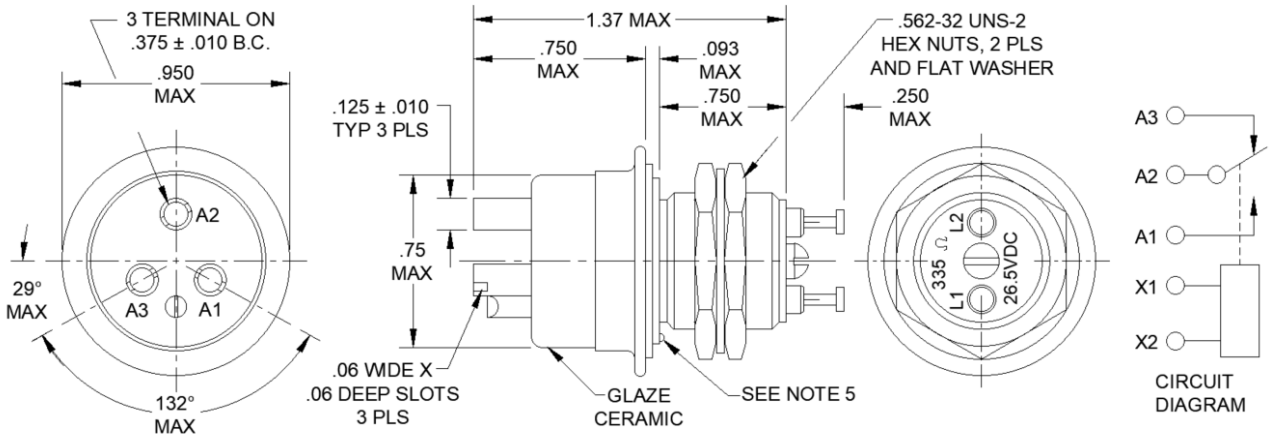
INCH-POUND
MIL-DTL-83725/4G
30 June 2022
SUPERSEDING
MIL-R-83725/4F(USAF)
15 April 1994

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPDT, 0.25 AMPERES, DC LOAD SWITCHING 2.5 KILOVOLTS DC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83725.



Inches	mm
.020	0.51
.030	0.76
.060	1.52
.083	1.65
.125	2.36
.250	6.35
.520	13.21
.750	19.05
.950	24.13
1.37	34.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Part or Identifying Number (PIN) shall be permanently marked on the relay envelop.
5. A .030 X .060 X .020 maximum rectangular index slot shall be used to accommodate the index pin.

FIGURE 1. Relay outline and dimensions.

AMSC N/A FSC 5945

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REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT.

Arrangement: One form C.

Load ratings:

Resistive load switching:

2.5 kV dc, 0.25 amperes.

Relay should be placed on ground side of load to prevent damage,

Contact resistance:
Rated life:
Before: .030 ohm, maximum.
During: .100 ohm, maximum.
After: .100 ohm, maximum

Contact bounce: Not applicable.

Capacitance:

2 picofarads, maximum across open contacts.
2.5 picofarads, maximum between open contacts and ground.

COIL DATA: At +25°C.

Duty rating: Continuous.

Maximum voltage: See table I.

Nominal voltage: See table I.

Pickup voltage: See table I.

Dropout voltage: See table I.

Coil resistance: See table I.

Operate time: 8 milliseconds, maximum over the temperature range (or measured using 18.8 V dc coil voltage at 25° C).

Release time: 8 milliseconds, maximum over the temperature range (or measured using 18.8 V dc coil voltage at 25°C).

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ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure:

Between all mated contacts in the open position: 5 kV dc.
Between high voltage terminals and housing: 5 kV dc.
Between coil and housing: 500 V rms, 60 Hz.

At reduced barometric pressure:

In accordance with MIL-STD-202-105, test condition C. Relays shall be mounted by normal mounting means. Points of application: Same as "at atmospheric pressure" except test voltage applied shall be 350 volts rms with all terminals grounded. Following the test, relays shall be examined for evidence of arcing, flashover, insulation breakdown, and damage.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202- 204, test condition C, 10 g's, 55 to 2,000 Hz.

Shock: MIL-STD-202-213, test condition A, 50 g's, 11 milliseconds half sine.

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1 ounce, maximum.

Marking: Applicable (circuit diagram need not be marked on relay).

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TABLE I. Operating characteristics. 1/

Dash number	Maximum voltage	Nominal voltage	Pickup voltage	Dropout voltage	Coil resistance	Pickup voltage over temperature range	Dropout voltage over temperature range
-001 -002	32 V dc 18 V dc	26.5 V dc 15 V dc	14 V dc max 8 V dc max	1.5 - 7 V dc 0.7 - 4.5 V dc	335 ohms ±10% 110 ohms ±10%	22 V dc 12.5 V dc	1 - 10 V dc 0.5 - 6 V dc

1/ At +25°C.

LIFE TEST REQUIREMENTS: Resistive load switching life.

Voltage (kV dc)	Current (amperes)	Life (operations)
2.5	0.25	10^5

NOTE: Load life switching is not accumulative, and each load life rating is to be considered a complete load life rating is to be considered a complete load life specification.

QUALITY ASSURANCE:

Reliability:

Units are to be run 50,000 cycles mechanical, loads shall be 40 milliamperes at 12 V dc. No miss allowed.

Seal:

Parts shall be stored for 20 days and retested for breakdown at 5 kV dc (for qualification only).

Dielectric withstanding voltage (for group A testing):

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5 to 10 seconds at a 10 percent increase in the dielectric withstanding voltage.

PIN: M83725/4 (and a dash number from table I)

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:
MIL-STD-202-204 MIL-STD-202-213 MIL-STD-202-105

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Custodians:
Air Force – 85
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Preparing activity:
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(Project 5945-2022-033)

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